

Problem Set 6 (due 11:59 pm, 1 December 2025)

Portfolio Management with Principal Component Analysis

This problem should be completed with Python.

(1) Download from yfinance the daily adjusted closing price of the following 10 S&P 500 stocks with the largest capitalization:

NVDA, AAPL, MSFT, GOOGL, AMZN, AVGO, META, TSLA, BRK-B, JPM

The period is from 1 August 2024 to 31 October 2025. (In Yahoo!Finance you need to type the end date as 11/1/2025.) Calculate the return $R_i(t)$ of the stock i on day t from 1 August 2024 to 31 October 2025 using the formula

$$R_i(t) = \ln x_i(t+1) - \ln x_i(t),$$

where $x_i(t)$ is the stock price on day t .

In (2), (3) and (4) below, please display the numerical values rounded to 3 decimal places, so that the matrix structure can be visualized. However, please keep the unrounded numerical values for accurate subsequent computations.

Remark: Above the print statements, you may add the Python line

```
with np.printoptions(precision=3) :
```

(2) Based on the data from 1 August 2024 to 31 July 2025, construct and output the correlation matrix of the 10 stocks.

(3) Compute and output the eigenvalues and eigenvectors of the correlation matrix, respectively.

(4) Sort the eigenvalues and eigenvectors in descending magnitude of the eigenvalues. Output the sorted eigenvalues and eigenvectors.

Remark: The Python command `np.argsort` is useful. Furthermore, to sort the eigenvalues in descending order, add `::-1` at the end of the command.

Remark: When sorting the eigenvectors, please pay attention to whether you correctly sort the columns rather than the rows.

(5) Plot the eigenvalues of the eigenmodes versus the eigenmode numbers. You may find that it is difficult to determine the number of principal portfolios to be included, probably due to the small number of stocks being studied. Hence, we will use a different criterion in the next part.

(6) Compute the cumulative variance of the eigenmodes. Let N be the number of the first few principal portfolios whose cumulative variance is highest below the 80% cutoff. Determine N .

Cluster Visualization

(7) Plot the 10 stocks in the space of the 2nd and 3rd principal components. Comment on any clusters that you observe, and comment on the nature of the clusters.

Rebalancing of Portfolios

In the following diversification strategies, the weights of the stocks may deviate away from the original ratios after a while due to different price movements of individual stocks. Rebalancing the portfolio is needed to bring it back to the original ratios. This involves selling overweight stocks and buying underweight stocks. Stocks in short positions are adjusted similarly. During rebalancing, the total amount of investment is not changed.

Diversification Strategies

Below we investigate 4 diversification strategies.

(A) Stock investment with equal weight

(8) In this strategy, the capital allocated to each of the 10 stocks is the same. Considering the stock price, compute the volume of stocks to be invested. Then compute the value of the portfolio during the period. Rebalance the portfolio on 2 February 2025.

(B) Risk Parity

(9) In this strategy, the risk of stock i per unit of capital is estimated by its standard deviation σ_i (note that the risk correlations of the stocks are neglected). The capital allocated to each of the 10 stocks is such that the risk of each stock is the same. Hence the capital allocated to each stock is proportional to $1/\sigma_i$. Compute the weight of the stocks to be invested. Then compute the value of the portfolio during the period. Rebalance the portfolio on 2 February 2025.

(C) Equal Weight Portfolio (or 1/N Strategy)

(10) In this strategy, the first N principal portfolios are selected, and equal weight is allocated to each of them. The first step is to determine the signs of the eigenvectors $\pm \mathbf{u}_a$ to be adopted. We will first calculate the projections of the principal eigenvectors to the equal-weight portfolio, which is the portfolio allocating equal capital to the 10 stocks. Compute the projections.

(11) The sign-corrected principal eigenvectors are those principal eigenvectors that have positive projections to the equal-weight portfolio. Compute the sign-corrected principal eigenvectors. Then compute the total weight of each stock contributed from the N principal portfolios.

(12) Which stocks in this strategy are in short positions? Compute the value of the portfolio during the period, including a rebalancing strategy on 2 February 2025. Note that the initial value is no longer equal to 1 due to the short-positioned stocks. You are required to normalize the initial value to 1.

(D) Diversified Risk Parity

(13) In this strategy, the capital allocated to each of the N principal portfolios are such that the risk of each portfolio is the same. Since the eigenvalue λ_a of each portfolio is the variance, the capital allocated to each principal portfolio is proportional to $1/\sqrt{\lambda_a}$. Compute the volume of stocks to be acquired. Which stocks in this strategy are in short positions? Then compute the value of the strategy during the period. Rebalance the portfolio on 2 February 2025.

To avoid confusion in implementing portfolio rebalancing, you are NOT required to normalize the initial value to 1.

Comparison of the Strategies

(14) Plot the daily values of the portfolios of the 4 strategies. Comment on any observations you make about the performance of the strategies.

(15) The ability of the 4 strategies to reduce risks is compared by calculating the following parameters:

Gain (the final value divided by the initial value)

Standard deviation

Minimum value (divided by the initial value)

Comment on any observations you make about the performance of the strategies.

(16) In practice, the real test of the performance of the strategies should be done with a different data set. Suppose investors adopt the portfolios prescribed by the 4 strategies constructed from 1 August 2024 to 31 July 2025 and start to invest during the period from 1 August 2025 to 31 October 2025. Compare the ability of the 4 strategies to reduce risks during this investment period.

For your further insights and interest, the names of the stocks are listed below:

NVDA	NVIDIA Corporation
AAPL	Apple Inc.
MSFT	Microsoft Corporation
GOOGL	Alphabet Inc. (Google)
AMZN	Amazon.com, Inc.
AVGO	Broadcom Inc.
META	Meta Platforms, Inc.
TSLA	Tesla, Inc.
BRK-B	Berkshire Hathaway Inc.
JPM	JP Morgan Chase & Co,